

- Standard palettes and blocks
- Model building and editing
- Sub-modelling for models reuse and simplification
- Customisation of built models
- Simulation includes a built-in Modelica compiler and C code generation for better performance

Results analysis and visualisation

One of the major advantages of this tool is that, unlike certain simulators, Xcos is a visual editor and, hence, much easier to pick up for a novice than a simulator that requires to learn coding in a particular language. There is no doubt that when compared to MATLAB/Simulink, it is not as polished. But, we must remember that the aim of Xcos is to introduce users to the world of simulation using a visual editor, and it is aimed mostly at students and academicians.

As mentioned earlier, simulation includes an integrated Modelica compiler and, hence, there is assurance of a legacy compiler for users' needs. It also supports batch simulation, cutting down on wait time for them.

The ability to customise a template means that a high degree of flexibility and, as a result, the ability to customise a solution is achieved. Make no mistake, this is a great tool for beginners, but you would be surprised as to how quickly you can level up on this software. I am speaking from personal experience; I did not understand the basics of Simulink at one point.

A further advantage lies in the fact that it is very easy to share models on Xcos, as it is easy to get access to the modelling environment. MATLAB/Simulink tends to bring out the cost factor a bit more, as all collaborators on a team need to have a licensed version, which can stretch most budgets. This is never an issue to take lightly and is offset only if the

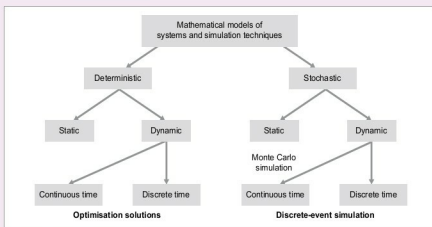


Fig. 3: Simulation systems

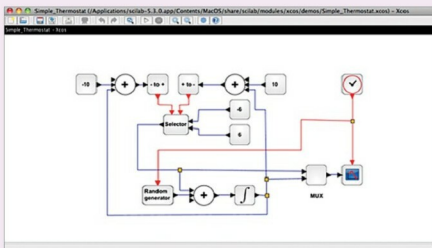


Fig. 4: Xcos screenshot

XCOS VS SIMULINK BY CRAIGLOUIE

	Matlab/Simulink/SimPowerSystems	Scilab/Xcos
Capability	Very good	Good
Documentation	Very good	Fair
User support	Very good	Fair
Installation	Fair	Fair
Sharing	Fair	Very good
Price	US\$ 12,000 + yearly fee	Free

licence is bought in bulk.

Final verdict

Regardless of other factors, if you are starting out in the area of visual editing and want software to help you ease into it, you should give Xcos (bundled in this month's *Electronics For You Plus*) a try. Its documentation is extensive, and you

can start off by trying out some of the simplest tutorials available. It would help to know your differential equations and your Kirchhoff's laws, but more than that, all you need is an inquisitive mind. **EFY**

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